



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
UNIT TEST
07-08-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TOAS]

Time : 3 Hours

Maximum Marks : 204

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **20** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.

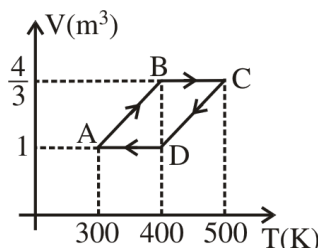
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. 2 mole diatomic gas undergoes cyclic process ABCDA as shown in indicator diagram. Then select correct alternative :-



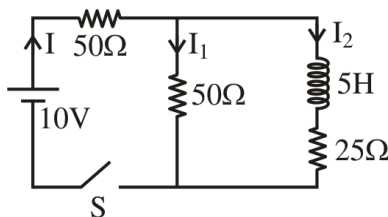
(A) Process AB, CD are isobaric and process BC, DA are isochoric.

(B) Work done in a cycle is $100R \left\{ 5 - \ln \left(\frac{16}{9} \right) \right\}$.

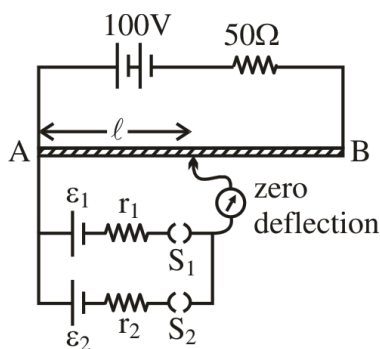
(C) Change in entropy of gas and surrounding is zero in a cycle.

(D) Change in entropy of gas in process CD is $R \ln \left\{ \frac{(3)^2 (4)^3}{(5)^5} \right\}$.

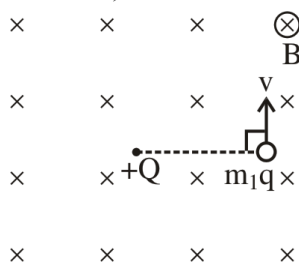
2. At $t = 0$ key S is closed then



- (A) Just after closing key $I_2 = 0$ and $I_1 = I = 0.1$ A.
 (B) Long after closing key $I = 0.15$ A, $I_1 = 0.05$ A & $I_2 = 0.1$ A.
 (C) Time constant of growth circuit is 0.1 s.
 (D) Time constant of growth circuit is 0.075 s.
3. Potentiometer wire AB has length 10m and resistance 50Ω . When only key S_1 is closed balanced length is 2m, when only key S_2 is closed balance length is 3m but when both keys are closed then balanced length is 2.5 m. Select correct alternatives.

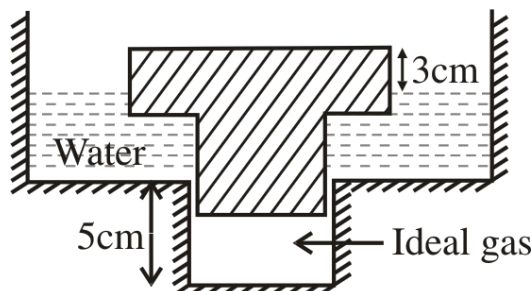


- (A) $\varepsilon_1 = 10$ V
 (B) $\varepsilon_2 = 15$ V
 (C) $r_1 = r_2$
 (D) Potential gradient on wire = 5V/m
4. A $+Q$ charged is fixed on a smooth horizontal table. A particle of mass m and charge q is placed at distance r from $+Q$ charge at given speed v . Uniform magnetic field B is inside plane of paper as shown. ($q/m = 10^{-2}$ C/kg, $Q = 10^{-6}$ C, $r = 10$ m, $B = 100$ T)

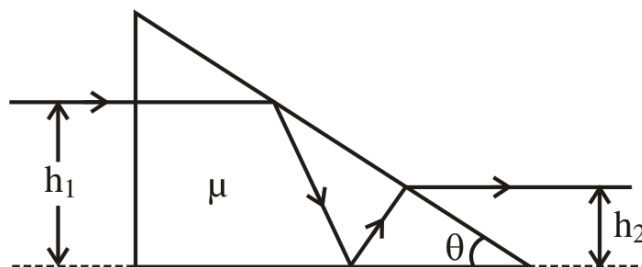


- (A) For $v = 1$ m/s, particle will move on a circular path of radius 10m with constant speed.
 (B) For $v = 0.9$ m/s particle will moving along a line with constant speed.
 (C) For $v = 5$ m/s, when particle reaches a distance 5m from $+Q$ it's speed becomes $\sqrt{7}$ m/s.
 (D) For $v = 9$ m/s, particle will move on a circular path of radius 10m with constant speed.

5. A cylinder of radius 10cm is filled with ideal gas and is filled with a piston made by connecting two solid cylinder of same thickness 5cm but different radius 10cm and 20cm as shown in figure. Initially when 3cm of piston is above water, only 2cm length of piston is inside cylinder, atmospheric pressure is 10^5 Pa and mass of piston is 5kg. Select correct alternatives.

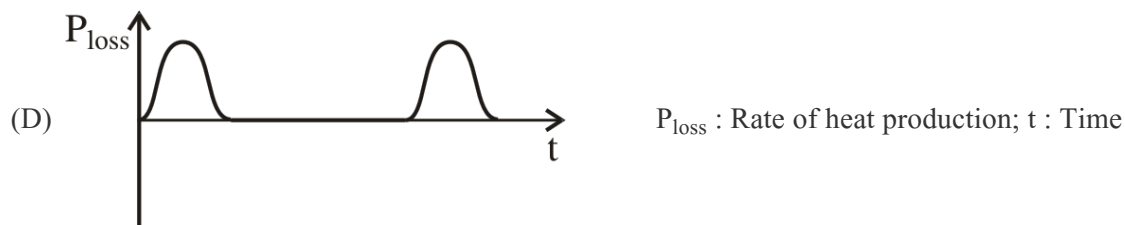
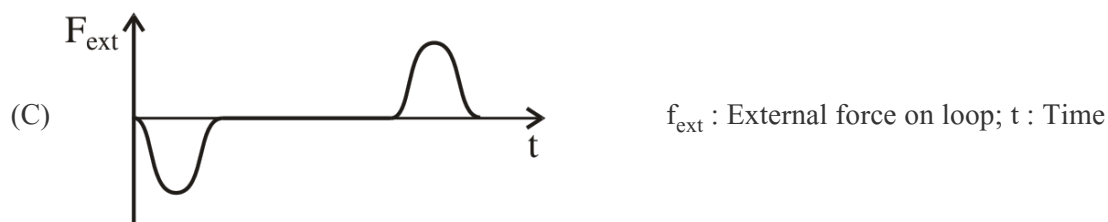
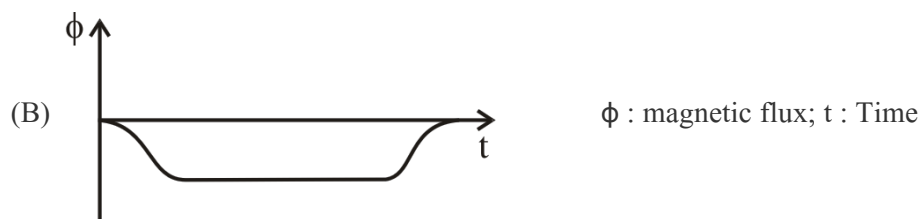
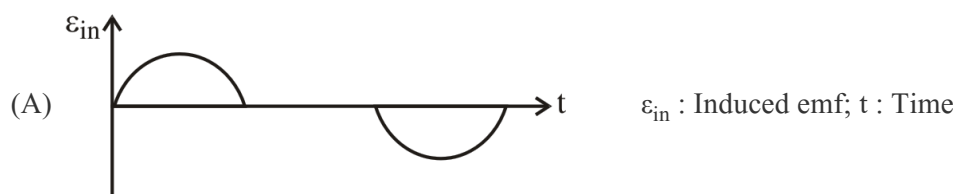
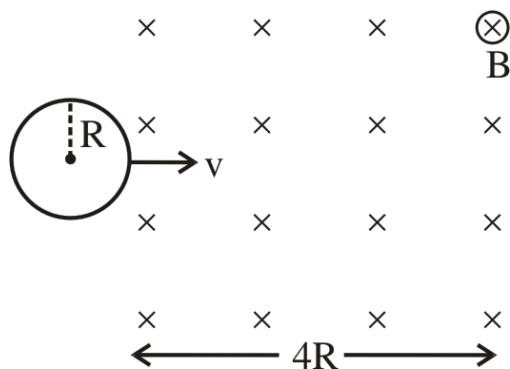


- (A) Initial pressure of ideal gas is less than 10^5 Pa.
- (B) Initial pressure of ideal gas is more than 10^5 Pa.
- (C) If water level increased by 2cm pressure of ideal gas will remain same.
- (D) If water level is increased by 2 cm pressure of gas will decrease.
6. When a ray parallel to base of right angled prism incident as shown in figure after two total internal reflection it emerge parallel to incident ray then :



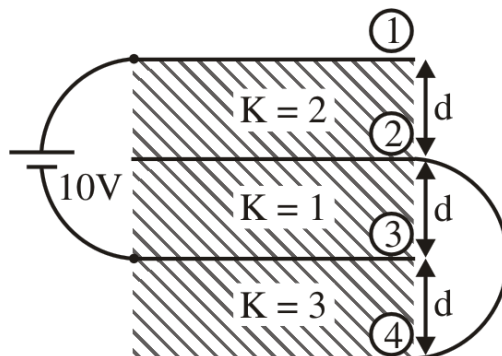
- (A) Refractive index of prism is $\frac{\cos \theta}{\cos 3\theta}$.
- (B) Refractive index of prism is $\frac{\sin 3\theta}{\sin \theta}$.
- (C) $\frac{h_2}{h_1} = \frac{\sin \theta}{\sin 3\theta}$
- (D) $\frac{h_2}{h_1} = \frac{\cos 3\theta}{\cos \theta}$

7. A conducting ring of radius R is moving with constant velocity v , at $t = 0$ it start entering region containing uniform magnetic field B inside plane of paper then select correct alternatives.



8. Four plates of equal area A and plate separation d are arranged as shown and filled with different

dielectric having dielectric constant K . $\left(\frac{\epsilon_0 A}{d} = 1\mu\text{F}\right)$ Select correct alternative.



(A) Effective capacitance of system is $\frac{4}{3}\mu\text{F}$.

(B) Charge on 2nd plate is $-10\mu\text{C}$.

(C) Charge on 3rd plate is $+\frac{40}{3}\mu\text{C}$.

(D) Energy stored in system is $\frac{200}{3}\mu\text{J}$.

SECTION-II : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

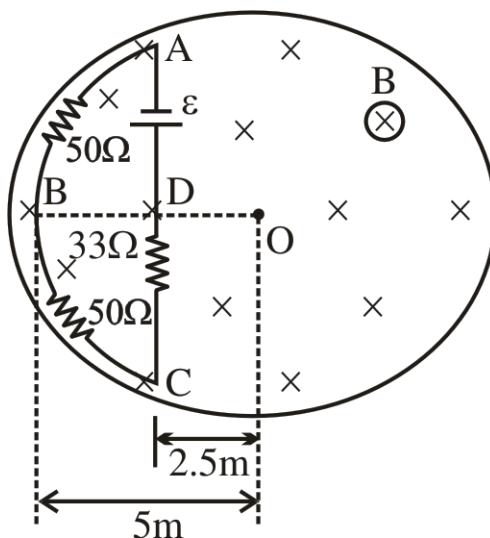
Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

In a cylindrical region uniform magnetic field (inside plane of paper) is increasing at constant rate

$(4\pi + 3\sqrt{3}) \frac{T}{s}$. A conducting loop ABCDA is placed as shown. ABC is arc with centre at axis of

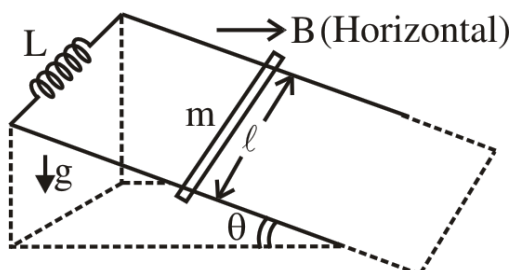
cylinder. Side AC also has a cell of EMF, $\varepsilon = 133/12$ volt, then : ($\pi^2 \approx 10$)



1. Electric current in loop will be (in A)
2. If potential difference between points C and D is x volt then $\frac{x}{100}$ is (take $4\pi\sqrt{3} \approx 22$)

Paragraph for Questions 3 and 4

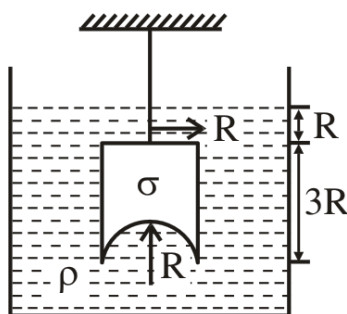
Two parallel conducting rails are inclined at angle θ from horizontal. Their upper ends are connected by an inductor coil of inductance L . A metallic rod of mass m starts sliding down. The rail and a uniform magnetic field B acts as shown. ($\theta = 30^\circ$, $L = 2H$, $m = 2kg$, $\ell = 1m$ and $B = 2T$, $g = 10 m/s^2$)



3. Time (in s) at which rod will lose contact with rail :
4. Velocity of rod (in m/s), just before it lose contact with rails :

Paragraph for Questions 5 and 6

A hemisphere of radius R is removed from a cylinder of radius R and length $3R$. This cylinder is made of material having density $\sigma = 2\rho$ and it is submerged in a fluid of density ρ as shown in figure with help of a string.



5. If tension in string is $x\rho(\pi R^3)y$ then value of x is -
6. If force applied by fluid on hemispherical surface is $y\pi R^3\rho g$ then value of y is -

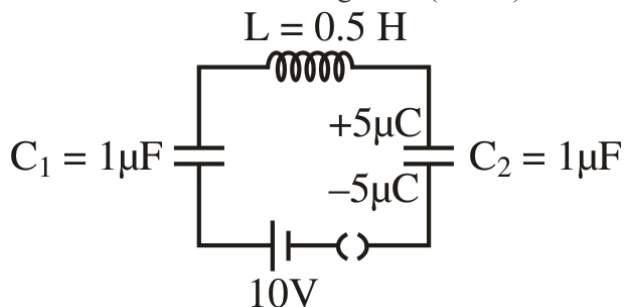
SECTION-III : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

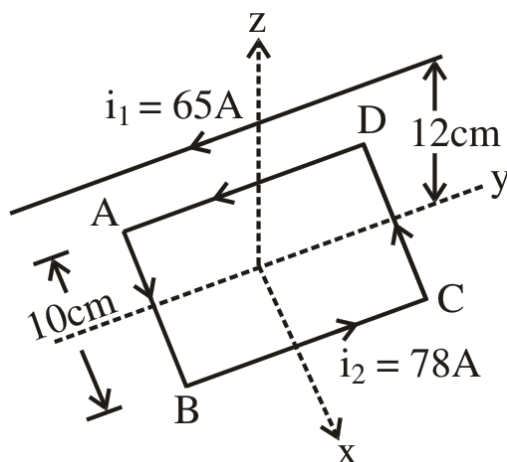
Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

1. At $t = 0$ key is closed then find maximum current through coil (in mA).

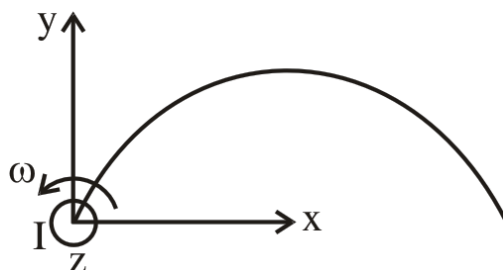


2. In an experiment an oil drop fall with constant speed v in a viscous fluid. When a downward uniform electric field is applied drop fall with constant speed $2v$. When uniform electric field αE is applied in upward direction than drop rises with constant speed v . Find α .
3. The self inductance of two coaxial long cylindrical shells of radius R and $2R$, carrying equal current along their length but in opposite direction is $L = \frac{\mu_0 \ell}{\alpha \pi} \ln(\beta)$, where ℓ is length of cylinder then $\alpha + \beta$ is
4. Figure shows a square loop 10cm on each side in x - y plane with its centre at the origin. An infinite long wire at $z = 12$ cm above y -axis.



If magnetic torque on loop is expressed as $x \times 10^{-7}$ Nm fill value of $x/90$.

5. Consider a rectangular block of wood moving with velocity $v_0 = 5\text{ m/s}$ in a gas at temperature 27°C and mass density 2 kg/m^3 . Velocity of block is perpendicular to its face having area $A = 1\text{ cm}^2$. Find drag force (in N) on the block of molar mass of gas is 10 g/mol $\left(R = \frac{25}{3} \frac{\text{J}}{\text{mol K}}\right)$.
6. Semi-circular conducting wire of radius $R = 1\text{ m}$, has its one end pivoted at origin and it is rotating about z-axis with constant angular velocity $\omega = 10\text{ rad/s}$. A current carrying long wire is along z-axis carrying current $I = 2\text{ A}$ in +z direction. Find potential difference across ends of wire.



PART-2 : CHEMISTRY
SECTION-I : (Maximum Marks: 32)

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1. A one litre solution contains 0.04 M Na_2HPO_4 and 0.02 M NaH_2PO_4 . For H_3PO_4 :
 $\text{pK}_{\text{a}_1} = 2.14$; $\text{pK}_{\text{a}_2} = 6.84$; $\text{pK}_{\text{a}_3} = 12.3$. The correct statement (s) is /are :

[Assume volume of solution is constant]

(use : $\log 2 = 0.3$, $\log 5 = 0.7$) :

- (A) pH of solution is 7.14
- (B) On adding 0.02 moles of HCl to initial solution pH increases by 0.6 units
- (C) On adding 0.02 moles of HCl to initial solution pH decreases by 0.6 units
- (D) On adding 0.01 mol NaOH to initial solution pH of solution becomes 7.54
2. Which of the following option(s) is/are correct ?
- (A) Angular functions of $2p_x$ and $3p_x$ orbitals in H-atom are identical
- (B) Radial functions of orbitals in H-atom mainly determine their energy and size
- (C) Net magnetic field of an atom is due to combined effect of orbital motion and spin of all its electrons in various shells
- (D) Radial functions of $2p_x$ & $2s$ orbitals are identical

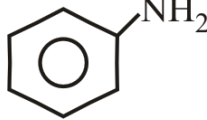
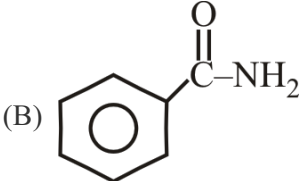
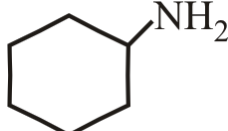
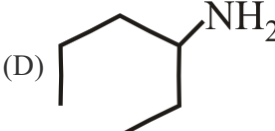
3. Which of the following match is/are correct?

- (A) Extraction Method - Self reduction
Metal(s) - Hg, Fe
- (B) Extraction Method - Electrolytic reduction
Metal(s) - Na, Al
- (C) Extraction Method - Hydrometallurgy
Metal(s) - Ag, Au
- (D) Extraction Method - Alumino-thermite reduction
Metal(s) - Mn, Cr

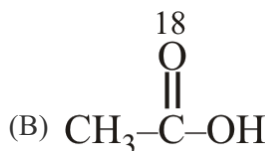
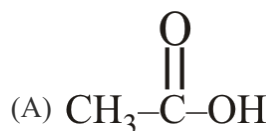
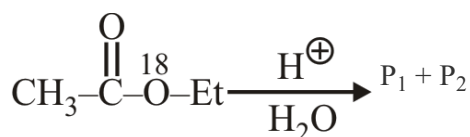
4. The correct statement(s) related to the metal extraction processes is(are)

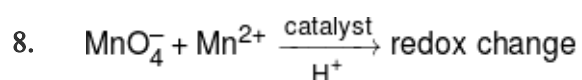
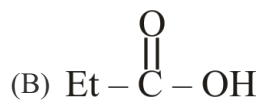
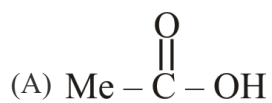
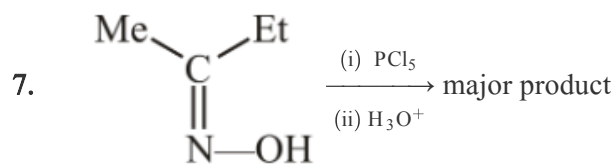
- (A) A mixture of PbS and PbO undergoes self-reduction to produce Pb and SO₂.
- (B) In the extraction process of copper from copper pyrites, silica is added to produce copper silicate.
- (C) Partial oxidation of sulphide ore of copper by roasting, followed by self-reduction produces blister copper.
- (D) In cyanide process, zinc powder is utilized to precipitate gold from Na[Au(CN)₂].

5. Which of the following compounds will produce a compound with offensive smell when treated with CHCl₃/KOH?

- (A) 
- (B) 
- (C) 
- (D) 

6. Which of the following will be the correct representation for products P₁, P₂ :





Select correct for above reaction.

(A) It is disproportionation reaction

(B) ZnO acts as catalyst

(C) ZnSO_4 acts as catalyst

(D) MnO_4^{2-} is one of the product

SECTION-II : (Maximum Marks: 12)

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- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

When 100 mL of 1.0 M HCl was mixed with 100 mL of 1.0 M NaOH in an insulated beaker at constant pressure, a temperature increase of 5.7 °C was measured for the beaker and its contents.

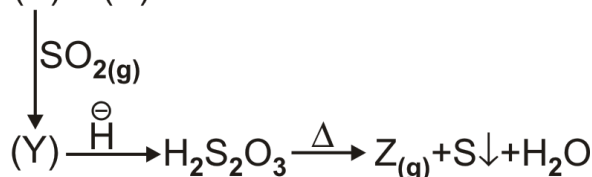
(Expt-1). Because the enthalpy of neutralisation of a strong acid with a strong base is a constant

(−57.0 kJmol^{−1}), this experiment could be used to measure the calorimeter constant. In a second experiment (Expt-2), 100 mL of 2.0 M acetic acid ($K_a = 2.0 \times 10^{-5}$) was mixed with 100 mL of 1.5 M NaOH (under identical conditions to (Expt-1)) where a temperature rise of 8.25 °C was measured.

(Consider heat capacity of all solutions as 4.2 Jg^{−1}K^{−1} and density of all solutions as 1.0 g mL^{−1})

1. Enthalpy of dissociation (in kJ mol^{−1}) of acetic acid obtained from the Expt-2 is
2. The pH of the solution after Expt-2

Paragraph for Questions 3 and 4

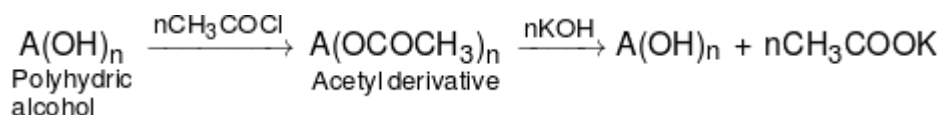


3. Difference in average oxidation number of S-atom present in anionic part of X and Y. (answer the magnitude value)
4. When Y is made to react with KI_{3(aq.)} solution then the average oxidation number of S-atom present in product will be.

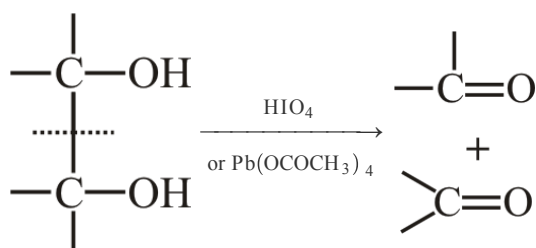
Paragraph for Questions 5 and 6

In a polyhydric alcohol, the total number of hydroxyl groups are estimated by the following methods:

(i) A known mass of the acetyl derivative of polyhydric alcohol is taken. It is mixed with an excess volume of a standard alkali solution. Both are refluxed. Alkali is used in the hydrolysis of acetyl derivative. The unused alkali is measured by doing titration with a standard acid solution and total number of hydroxyl groups (n) can thus be calculated stoichiometrically.



(ii) Total number of -OH groups in vicinal position can be calculated by periodic acid or lead tetra acetate oxidation.



5. 0.436 g of acetyl derivative of a polyhydric alcohol (molecular mass = 92) requires 0.336 g KOH for hydrolysis. The total numbers of hydroxyl group(s) in the alcohol are
6. A compound with molecular formula $\text{C}_4\text{H}_{10}\text{O}_3$ is converted by the action of acetyl chloride to a compound(s) with molecular weight 190. How many number of acetyl group present in compound (B)

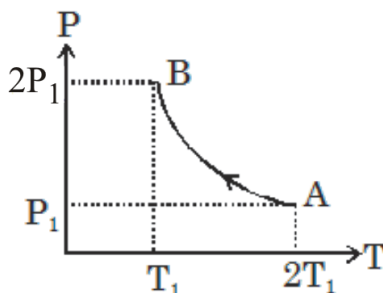
SECTION-III : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

- Number of crystal system having at least 2 interfacial angles equal, is x
Number of crystal system having none of the interfacial angles equal, is y
Number of crystal system having none of the axial length to be equal, is z
Find the value of $x - y + z$.
- Two moles of an ideal monoatomic gas is taken from state A to state B through a process AB in which the pressure and the temperature of the gas vary such that $PT = \text{constant}$. The process can be represented on a P-T graph as follows ($T_1 = 300 \text{ K}$) :



The magnitude of heat involved during process AB is (Give your answer after dividing by $300R$) :

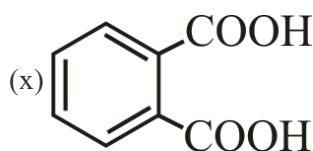
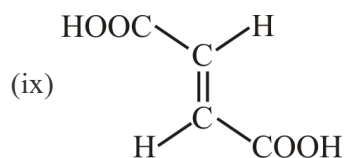
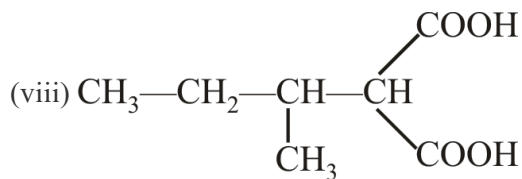
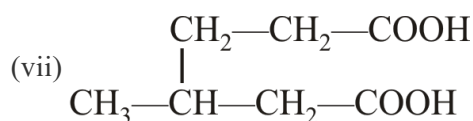
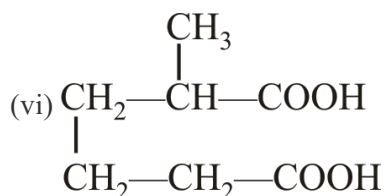
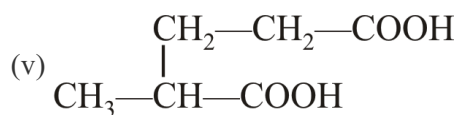
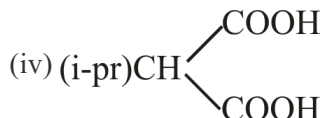
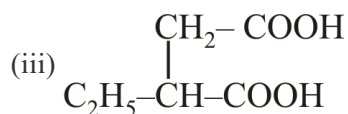
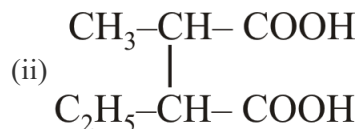
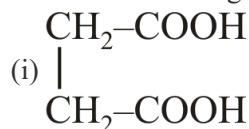
- When the formation of one mole HBr(g) takes place reversibly from gaseous H_2 and liq. Br_2 at 300K and 1 atm increase in entropy of the surrounding is 170 J/K . What is change in entropy (J/K) of the system if reaction $2\text{HBr(g)} \rightarrow \text{H}_2\text{(g)} + \text{Br}_2\text{(l)}$ is carried out reversibly at 300K in rigid container. [Given : $R = 8.3 \text{ J/mol-K}$]
Fill your answer as sum of digits (excluding decimal places) till you get the single digit answer.
- Find number of process or ore given in Column-I are correctly represented with Column-II. You have to consider P for P, Q for Q, R for R etc.

(I) Process/Ore		(II) Related to	
(P)	Matte formation	(P)	Cu
(Q)	Cynide leaching	(Q)	Au
(R)	Monds process	(R)	Reaction of metal + H_2
(S)	Fused oxide electrolysis	(S)	Al
(T)	Chromite ore	(T)	$\text{Fe} + \text{Cr}$
(U)	Van Arkel process	(U)	Zr
(V)	Sulphuric acid leaching	(V)	Low grade Cu ore
(W)	Separation of FeWO_4	(W)	Extraction of Pb

5. Number of compounds which on acidic hydrolysis gives carboxylic acid as one of the product :

- (i) CH_3CN (ii) CH_3NC (iii) CH_3CONH_2 (iv) CH_3COCl
 (v) $(\text{CH}_3\text{CO})_2\text{O}$ (vi) $\text{CH}_3\text{COOCH}_3$ (vii) CH_3CHO (viii) $\text{CH}_3\text{CH}_2\text{OH}$
 (ix) CH_3COCH_3 (x) CH_3OCH_3

6. Consider following carboxylic acids :



If :

(i) The number of given carboxylic acids which exhibit stereoisomerism & on heating makes cyclic anhydride is x ;

(ii) The number of given carboxylic acids which does not form any cyclic anhydride is y.

Report your answer as x + y?

PART-3 : MATHEMATICS
SECTION-I : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. $\int \frac{dx}{\sqrt{2} + \sin x + \cos x}$ is equal to -

(A) $\frac{1}{\sqrt{2}} \left(\sec \left(x + \frac{\pi}{4} \right) - \tan \left(x + \frac{\pi}{4} \right) \right) + C$

(B) $\frac{1}{\sqrt{2}} \left(\sec \left(x - \frac{\pi}{4} \right) - \tan \left(x - \frac{\pi}{4} \right) \right) + C$

(C) $\frac{1}{\sqrt{2}} \left(\tan \left(x + \frac{\pi}{4} \right) - \sec \left(x + \frac{\pi}{4} \right) \right) + C$

(D) $\frac{1}{\sqrt{2}} \left(\tan \left(x - \frac{\pi}{4} \right) - \sec \left(x - \frac{\pi}{4} \right) \right) + C$

2. A function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies the differential equation $2xy + (1 + x^2)y' = 1$, (where y' denotes derivative of y with respect to x) where $f(0) = 0$, then

(A) $f(x)$ is odd

(B) $f(x) = 1$ has no real solution

(C) $f(x) = \frac{1}{2}$ has exactly two solution

(D) $f(x) = -1$ has no real solution

3. Which of the following options is/are true ?
- (A) for $\alpha > 1$, $(1+x)^\alpha \geq 1 + \alpha x \forall x > -1$
 (B) for $\alpha > 1$, $(1+x)^\alpha < 1 + \alpha x \forall x > -1$
 (C) for $x > 0$, $e \ln x \leq x$
 (D) for $x > 0$, $e \ln x > x$
4. Area enclosed by parabola $ay = 3(a^2 - x^2)$ and x-axis is 64, then value of 'a' can be -
- (A) 4 (B) 2
 (C) -4 (D) -2
5. Let A be the area bounded by the curve $f(x) = \frac{2}{1+x^4}$, x-axis, y-axis and ordinate $x = 1$, then -
- (A) $A > \frac{\pi}{2}$ (B) $A > 2 \ln 2$
 (C) $0 < A < \frac{\pi}{2}$ (D) $\frac{\pi}{2} < A < 2 \ln 2$
6. Let C be a curve such that normal at any point P on it meets x-axis and y-axis at A and B respectively. If $BP : PA = 1 : 2$ (internally) and curve passes through (0,4), then -
- (A) The curve 'C' passes through $(\sqrt{10}, -6)$
 (B) Equation of tangent at $(4, 4\sqrt{3})$ is $\sqrt{3}x - 2y + 4\sqrt{3} = 0$
 (C) Equation of tangent at $(4, 4\sqrt{3})$ is $2x - \sqrt{3}y + 4 = 0$
 (D) If point P is (8, 12), then point A and point B are (24, 0) and (0, 18) respectively.
7. Let S be the area bounded by $y = e^{|\cos 4x|}$, $x = 0$, $y = 0$ and $x = \pi$. Then
- (A) $S = 2 \int_0^{\pi/2} e^{\sin t} dt$ (B) $S > \frac{\pi}{2}$
 (C) $S < 2 \left(e^{\frac{\pi}{2}} - 1 \right)$ (D) $S \leq 1$
8. The curve(s) which satisfy the differential equation $x dy = y dx + \sin\left(\frac{1}{x}\right) dx$ is/are -
- (A) $y = x \left(\cos\left(\frac{1}{x}\right) + 3 \right)$
 (B) $y = 2x \left(2\cos^2 \frac{1}{2x} + \sin^2 \frac{1}{2x} \right)$
 (C) $y = x^2 \left(\cos\left(\frac{1}{x}\right) + 3 \right)$
 (D) $y = 2x \left(2\cos^2 \frac{1}{2x} - \sin^2 \frac{1}{2x} \right)$

SECTION-II : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

Let $g(x) = \frac{x(x-1)}{(x^2+1)}e^{-x}$ and function $y = f(x)$ satisfy the differential equation $dy - ydx = e^x g'(x)dx$ also $f(0) = 0$

1. $f'(1)$ is equal to -
2. If area bounded by $f(x)$ and x -axis in IV quadrant is $\frac{\pi}{a} - \ln\left(\frac{e}{\sqrt{2}}\right)$, then a is equal to -

Paragraph for Questions 3 and 4

Consider a function $f(x) = \frac{x^3}{3} - \frac{x^2}{2} + x + \frac{1}{12}$.

3. The value of $\sum_{r=1}^{99} f\left(\frac{r}{100}\right)$ is -
4. The value of integral $\int_{1/7}^{6/7} f(f(x))dx$ is -

Paragraph for Questions 5 and 6

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x) = \max \{|x|, |x-1|, |x-2|, \dots, |x-2018|\}$$

$$g(x) = \min \{|x|, |x-1|, |x-2|, \dots, |x-2018|\}$$
 then :

5. The value of $\int_0^{2018} f(x) dx$ is $3(a)^2$ then value of a is :
6. The value of $\frac{\int_0^{2018} f(x) dx}{\int_0^{2018} g(x) dx}$ is

SECTION-III : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

1. Let $f : [0, 1] \rightarrow (0, \infty)$ be a continuous function such that $\int_0^1 f(x) dx = 1$, then maximum value of $\left(\int_0^1 \sqrt[3]{f(x)} dx \right) \left(\int_0^1 \sqrt[5]{f(x)} dx \right) \left(\int_0^1 \sqrt[7]{f(x)} dx \right)$, is
2. The value of $\sqrt{2} + \int_0^1 \frac{2x^2 - 2x + 1}{\sqrt{x^2 - 2x + 2}} dx$ is -
3. The value of $\int_1^3 \left(\sqrt{1 + (x-1)^3} + (x^2 - 1)^{1/3} \right) dx$, is
4. The curve $y = e^{2x} - 18x + 15$ crosses the y-axis at P and has point of minima at Q. If the area bounded by the curve and chord PQ is $(k \ln 3 - k + 1)$ then k is equal to (use $\ln 3 = 1.1$)
5. Let $a = \cos^{-1} \cos 20$, $b = \cos^{-1} \cos 30$, $c = \sin^{-1} \sin(a + b)$ and $5 \sec^{-1} \alpha + 10 \sin^{-1} \beta = 10(a + b + c)$ if the value of $\tan^{-1} \alpha + \cos^{-1} (\beta - 1)$ is k then value of $\left(\frac{12}{\pi} k \right)$ is
6. If 'e' denotes the eccentricity of the hyperbola, satisfying the differential equation $2xy \frac{dx}{dy} = x^2 + y^2$ and passing through the point (2, 1), then $(e^2 - 1)$ is equal to

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

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